Predicting 10-Year Risk of Coronary Heart Disease

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Dataset Overview

id	age	education	sex	is_smoking	cigsPerDay	BPMeds
0	64	2.0	F	YES	3.0	0.0
1	36	4.0	M	NO	0.0	0.0
2	46	1.0	F	YES	10.0	0.0
3	50	1.0	M	YES	20.0	0.0
4	64	1.0	F	YES	30.0	0.0
5	61	3.0	F	NO	0.0	0.0
6	61	1.0	M	NO	0.0	0.0

Content:

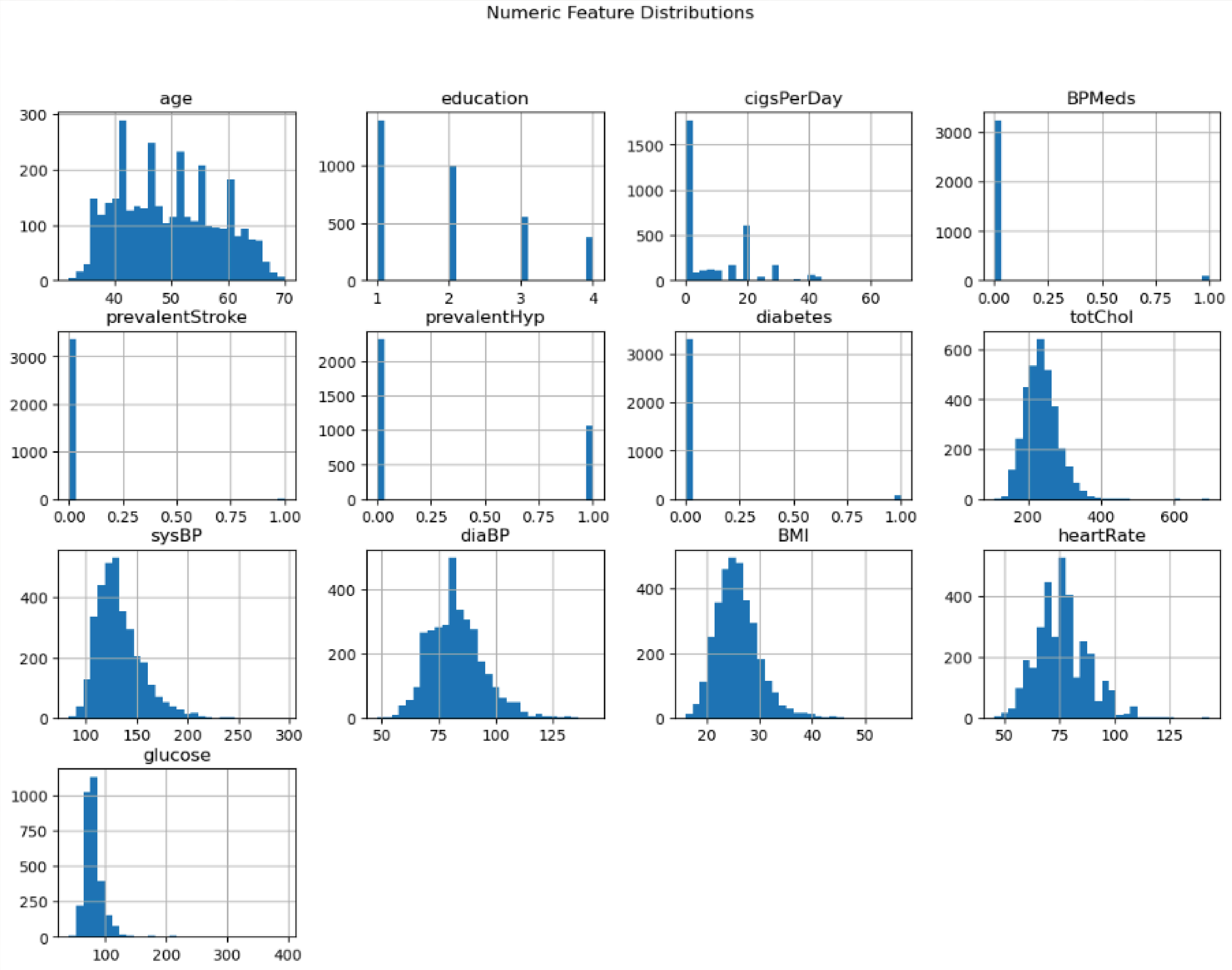
- 3,390 patient records
- 17 variables
- Target: TenYearCHD (0 = No, 1 = Yes)
- CHD prevalence ≈ 15%

Data Quality & Preprocessing

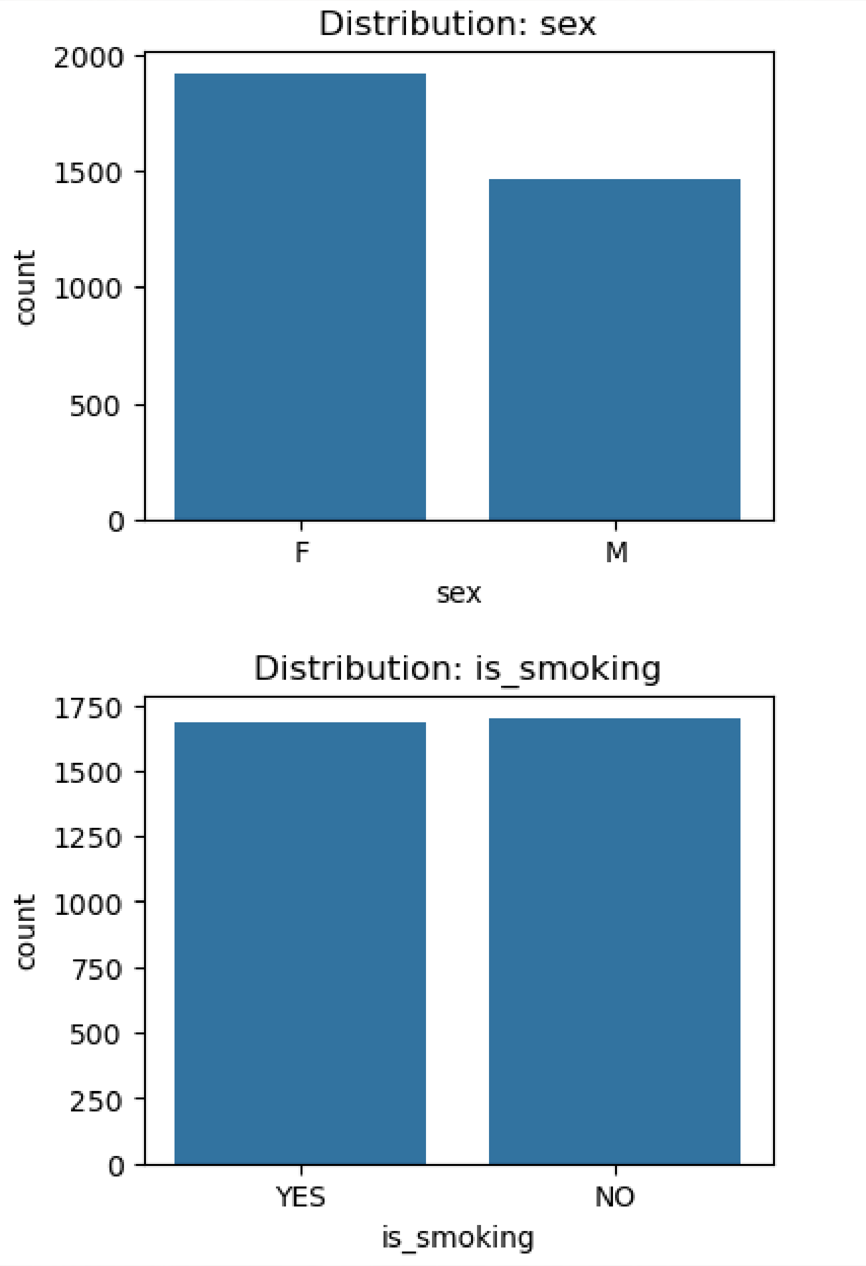
- Missing values in glucose, BMI, cholesterol
- Median imputation for numeric features
- One-hot encoding for categorical variables
- Feature scaling applied

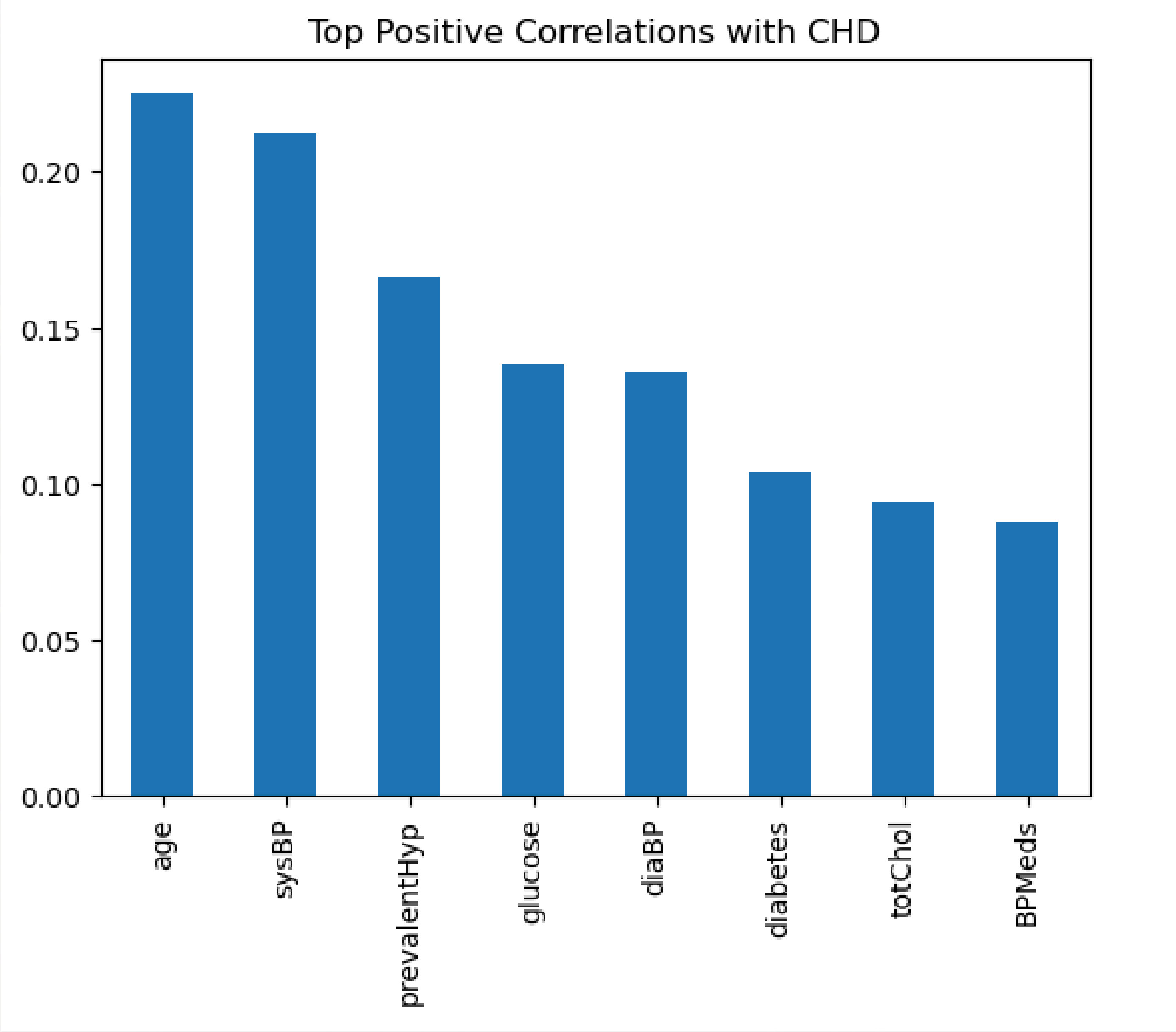
Exploratory Data Analysis

- Strong imbalance in CHD cases
- Age and systolic BP show strongest correlation
- Smoking and diabetes increase CHD risk
- Several features are right-skewed



Exploratory Data Analysis





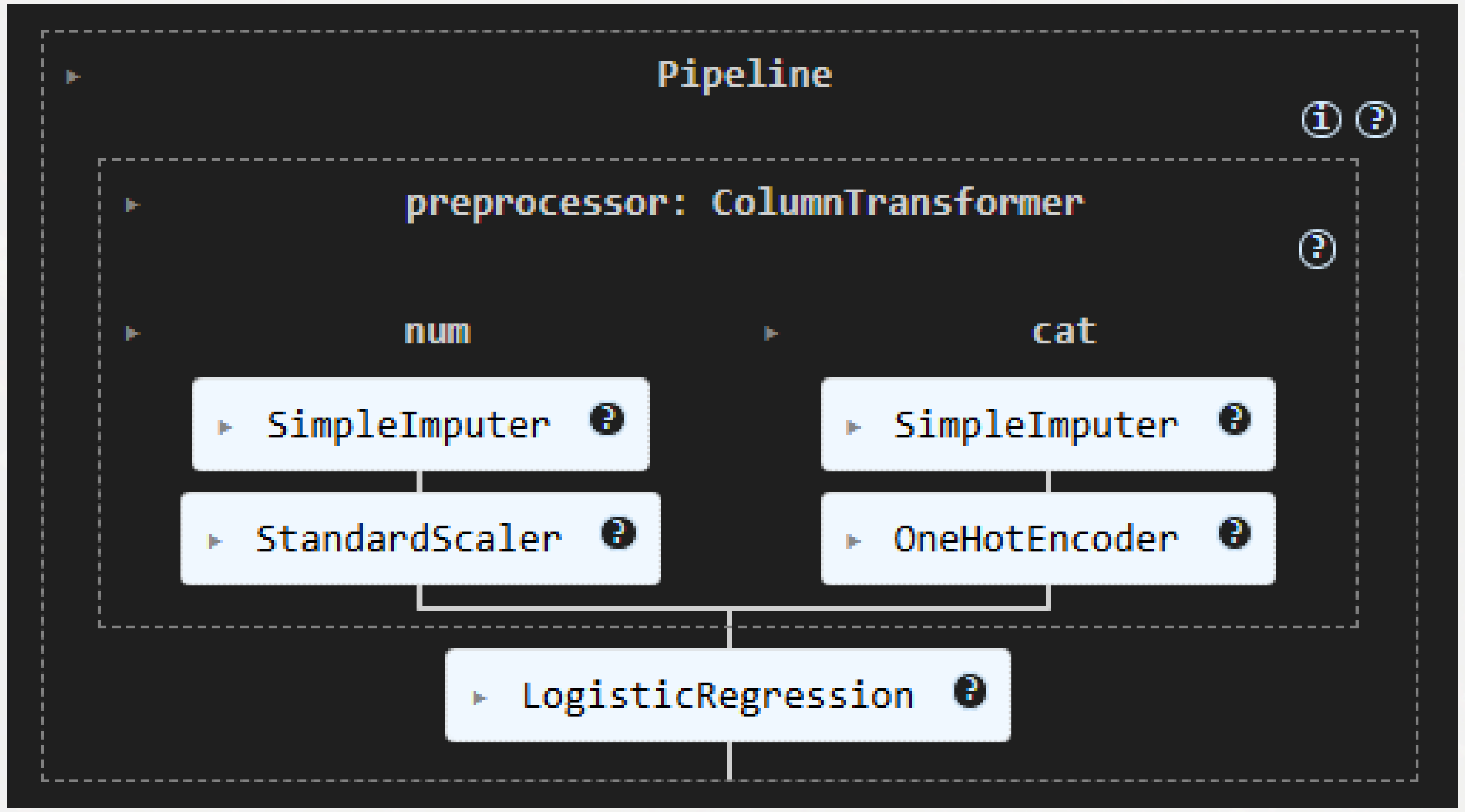
TenYearCHD	0	1
sex		
F	0.875715	0.124285
M	0.814588	0.185412

is_smoking

TenYearCHD	0	1
is_smoking		
NO	0.861421	0.138579
YES	0.836989	0.163011

Model & Methodology

- Train/Test split with stratification
  Training set: 2542 rows
  Test set: 848 rows
- Pipeline used for preprocessing + modeling



Model Performance

- ROC AUC ≈ 0.74
- Recall (CHD) ≈ 70%
- Accuracy ≈ 72%
- Accuracy not sufficient due to imbalance
- Optimal threshold chosen via F1-score
- Balanced precision vs recall

```
[[525 195]
 [ 39  89]]
              precision    recall  f1-score   support

           0       0.93      0.73      0.82       720
           1       0.31      0.70      0.43       128

    accuracy                           0.72       848
   macro avg       0.62      0.71      0.62       848
weighted avg       0.84      0.72      0.76       848
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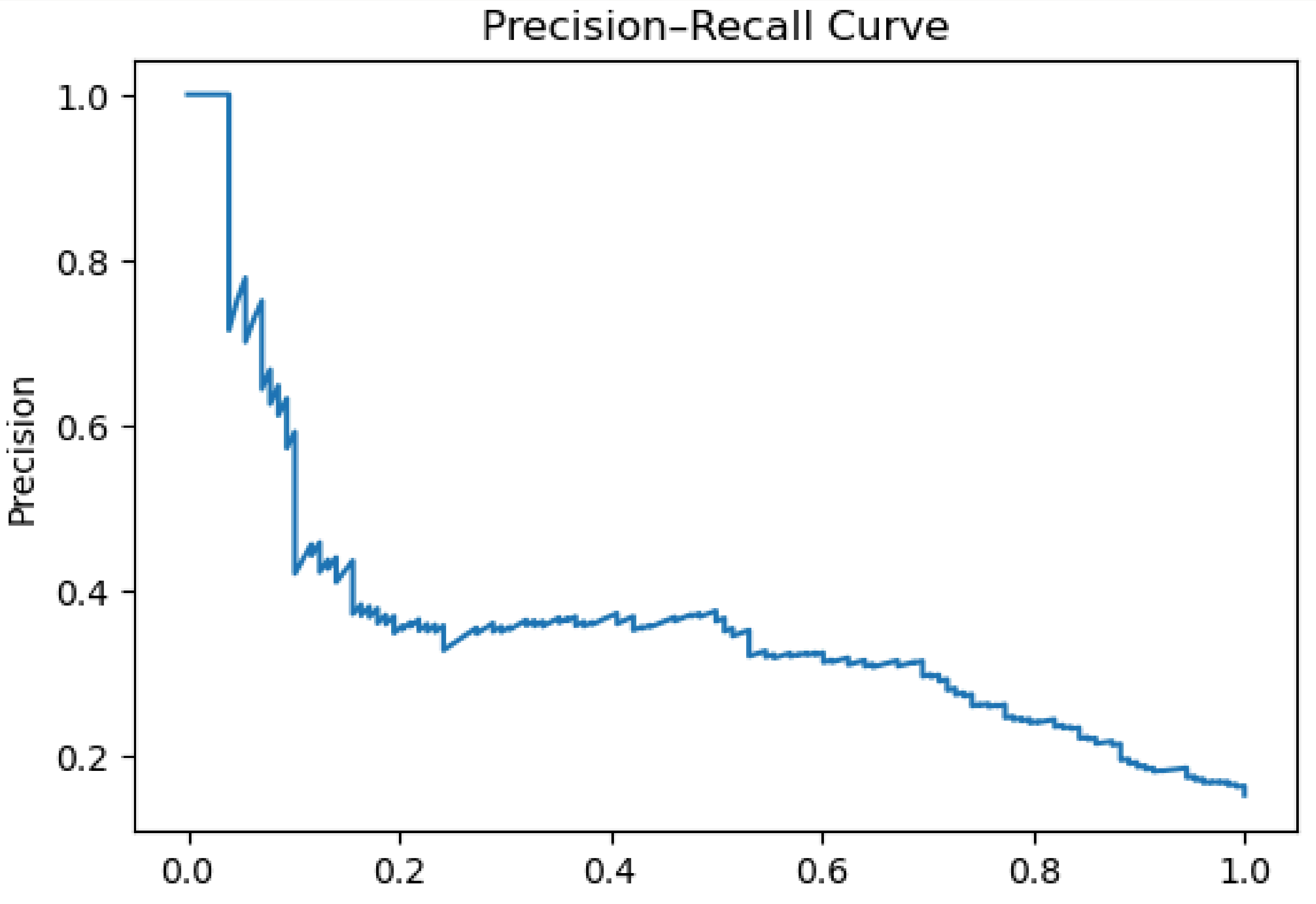

Insights & Improvements

Insights:

- Age and blood pressure are key drivers
- Lifestyle factors matter

Improvements:

- Handle non-linear effects
- Address class imbalance further
- External validation

The End

THANK YOU FOR LISTENING